

Pow3R: Empowering Unconstrained 3D Reconstruction with Camera and Scene Priors

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Figure 1. **Qualitative comparison** of raw 3D reconstructions between DUST3R and Pow3R for the image pair on the left. DUST3R cannot leverage auxiliary information potentially available at test time, resulting in noisy and low-resolution estimates, whereas Pow3R allows to inject such priors at test time in a versatile manner, enabling new capabilities such as high-resolution processing as a direct by-product.

Abstract

We present Pow3R, a novel large 3D vision regression model that is highly versatile in the input modalities it accepts. Unlike previous feed-forward models that lack any mechanism to exploit known camera or scene priors at test time, Pow3R incorporates any combination of auxiliary information such as intrinsics, relative pose, dense or sparse depth, alongside input images, within a single network. Building upon the recent DUST3R paradigm, a transformer-based architecture that leverages powerful pre-training, our lightweight and versatile conditioning acts as additional guidance for the network to predict more accurate estimates when auxiliary information is available. During training we feed the model with random subsets of modalities at each iteration, which enables the model to operate under different levels of known priors at test time. This in turn opens up new capabilities, such as performing inference in native image resolution, or point-cloud completion. Our experiments on 3D reconstruction, depth completion, multi-view depth prediction, multi-view stereo, and multi-view pose estimation tasks yield state-of-the-art results and confirm the effectiveness of Pow3R at exploiting all available information. The project webpage is <https://europe.naverlabs.com/pow3r>.

1. Introduction

Building general models for 3D perception that can unify different 3D vision tasks such as depth estimation, key-point matching, dense reconstruction or camera pose prediction, remains a complex and exciting open challenge. Traditional Structure from Motion (SfM) approaches such as COLMAP [59] do not apply pre-learned priors, so each scene must be optimized independently. On the other hand, most learning-based approaches focus on specific 3D vision tasks such as depth completion [26, 52], monocular and stereo depth estimation [46], pose estimation [73, 89] or novel view synthesis [27, 42]. While these dedicated approaches have been successful to some extent, this trend contrasts with other domains such as natural language processing where a single model like ChatGPT [45], trained with a vast collection of data, solves a range of different tasks without requiring re-training. While some models like CroCo-v2, DINOv2 or Stable Diffusion [15, 54, 77], trained on large image datasets, have served as learned priors for other tasks, including 3D vision ones, they are still far from multi-purpose foundation models like ChatGPT is for NLP.

DUST3R [75] marked a significant step forward in regression-based 3D foundation models. DUST3R performs

3D dense reconstruction from uncalibrated and unposed images, solving many downstream vision tasks at once without the need for fine-tuning. Its success can be attributed to two main factors: (i) its use of a pointmap representation, which provides flexibility over its output space and enables to regress camera parameters, poses and depth easily and (ii) its use of a transformer architecture that exploits powerful pretraining combined with a vast amount of 3D training data. Despite its huge success, DUST3R [75] and its successor MAST3R [31] are extremely limited in terms of their input space, *i.e.* they only take RGB images. Yet, in practice, many real-world applications *do* provide extra modalities such that calibrated camera intrinsics, sparse or dense depth from RGB-D sensors or LIDAR, *etc.* This raises the question of how best to exploit additional input modalities, and whether this not just boosts performance but also enables new capabilities, such as obtaining the full depth from sparse point clouds of SfM or LiDAR.

In this paper, we introduce Pow3R, a new 3D feed-forward model that empowers DUST3R with any subset of priors available at test time such as camera intrinsics, sparse or dense depth, or relative camera poses. Each modality is injected into the network in a lightweight fashion. To allow the model to operate under different conditions at test time, random subsets of input modalities are fed at each training iteration. As a result, we obtain a single model that performs on par with DUST3R when no prior information is available but outperforms it when it exists. Our model also gains new capabilities as a by-product: for instance, the camera intrinsics input allow to process images whose principal point is far from the center, thus allowing to perform extreme cropping, *e.g.* for performing sliding window inference. As a side contribution, our model directly outputs the pointmaps of the second image in its coordinate system, allowing faster relative pose estimation. To sum up, our contributions are:

- Pow3R is a holistic 3D geometric vision model capable of taking any subset (including none) of camera intrinsics, pose and depthmaps with corresponding input images.
- Extensive experiments demonstrate an important boost in performance over DUST3R which cannot exploit priors, yielding state-of-the-art results in three benchmarks.
- By predicting the same pointmaps in two different camera coordinate systems, we can achieve more accurate relative pose, orders of magnitude faster.

2. Related work

Structure-from-Motion and MVS. SfM [12, 13, 22, 25, 59] builds 3D sparse maps by optimizing camera parameters using only RGB images. Traditional approaches consists of handcrafted pipelines to match keypoints [5, 7, 21, 36, 55], form pixel correspondences across multiple images, determine geometric relations, and perform bundle

adjustment to optimize camera locations. Recently, SfM has incorporated learning-based approaches leveraging neural networks as components [4, 10, 35, 57, 66, 70, 74, 85], but it remains brittle to outliers and challenging conditions. Likewise, multi-view stereo (MVS) aims to reconstruct dense 3D surface through handcrafted pipelines based on triangulation from multiple viewpoints [18, 19, 60]. Learning-based approaches have also been incorporated for MVS to reconstruct 3D from few images [24, 30, 44, 64, 78, 83], and more recently by adopting the RAFT architecture [40] or using generic Transformers [9]. In contrast, the recent DUST3R framework [31, 75], upon which we build our approach, departs from these methods altogether by defining a simple and unified neural framework to 3D reconstruction based on the paradigm of pointmap regression. The idea is that, ideally, learning-based MVS should be capable of reconstructing 3D from a few images without strictly requiring the availability of camera parameters.

Guiding 3D. In this paper, we refer to *guidance* as the notion that a method can *optionally* take in 3D priors (typically camera parameters or sparse depth) to help with some prediction task. This can take different forms in the vast literature of 3D vision works. COLMAP [60], for instance, can take camera parameters for rectifying camera views; however, it is an optimization-based method, meaning that each scene has to be processed independently. Recently, UniDepth [46] is a monocular depth and intrinsic estimation method capable of optionally providing camera intrinsics as additional guide to generate depthmap, with an architecture specifically handcrafted for that purpose. Also, depth completion models [68, 69, 76, 80] such as CompletionFormer [26] take sparse depth input as guidance in order to inpaint the full depthmap, but are also typically capable of inferring depth from an image alone. Overall, however, there is currently no learning-based model to the best of our knowledge that guides its output by taking any subset of sparse depth, camera intrinsics and camera extrinsics.

Depth Completion. Given sparse depths and a corresponding RGB image, the task is to densify depth by propagating sparse measurements. CNN-based approaches were adopted by [38, 39], and [11] proposed convolutional spatial propagation network (CSPN). Multi-branch networks, which process RGB and sparse depth separately, were used in [23, 43, 48, 52, 68, 71, 79, 90]. CompletionFormer [26] leverages both CNN and ViT architectures to capture both global and local features. Most methods focus only on the specific depth completion task from a single image and partial depth as the input, often without incorporating camera intrinsics. Using the pointmap representation, Pow3R, based on a ViT architecture, can complete the depth map from both RGB images and sparse point clouds, while taking relative pose or camera intrinsics as auxiliary input.

Pose Estimation from Sparse views. Learning-based ap-

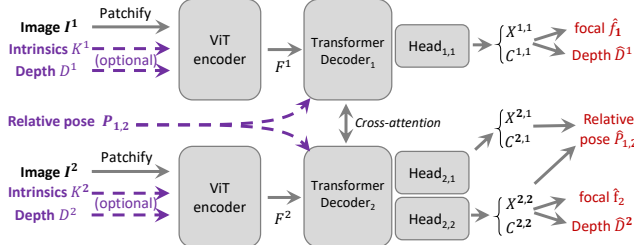


Figure 2. **Overview of the model architecture.** Following DUST3R [75], images are encoded then decoded with a ViT backbone into pointmaps, from which focals, depthmaps and relative pose can be extracted. Pow3R introduces optional inputs to guide the regression with prior knowledge about the camera intrinsics and depth (fed into the encoder) and the pose (into the decoder).

proaches involve regressing relative pose, applying probabilistic methods, using energy-based or diffusion models [3, 33, 53, 73, 88]. RayDiffusion [89] models camera rays instead of camera parameters, applying diffusion on rays, then recovers the camera pose using DLT [2]. DUST3R discovers camera pose from pointmaps using PnP, while Pow3R predicts pointmaps in both camera coordinate frames, allowing the use of Procrustes alignment.

DUST3R and its developments. DUST3R [75] does not require calibrated images for 3D reconstruction, can handle extreme viewpoint changes, and deals with both single and a few images within the same framework. Its pointmap representation eases the training by not having to deal with camera parameters directly. MAST3R [31] improves DUST3R by also predicting dense local features, trained with an additional matching loss. Splatt3R [65] marries MAST3R with GS. Spann3R [72] predicts pointmaps in global coordinate and adopts an external spatial memory that tracks relevant 3D information. MonST3R [87] further trains DUST3R on dynamic scenes and generates time-varying dynamic point clouds, as well as camera intrinsics and poses from a dynamic video. In this work, we take a different and complementary direction and instead improve the core capabilities of DUST3R, which could benefit all aforementioned works.

3. Methodology

We aim to train a network $\mathcal{F}$ that can take two input images $I^1, I^2 \in \mathbb{R}^{W \times H \times 3}$ of a given static scene and any subset of auxiliary information $\Omega \subseteq \{K_1, K_2, P_{1,2}, D^1, D^2\}$, in order to regress a 3D reconstruction of the scene. Here, $K_1, K_2 \in \mathbb{R}^{3 \times 3}$ are camera intrinsics, $P_{1,2} \in \mathbb{R}^{4 \times 4}$ denote the relative pose between the two cameras, and $D^1, D^2 \in \mathbb{R}^{W \times H}$ are depthmaps with associated masks $M^1, M^2 \in \{0, 1\}^{W \times H}$ specifying pixels with valid depth data (*i.e.* masks are possibly sparse). The network $\mathcal{F}$ is tasked to regress several pointmaps from which the camera intrinsic and extrinsic parameters as well as the dense depthmaps can be straightforwardly extracted for both im-

ages as described below.

Pointmaps. For each pixel (i, j) in an image I , we assume there exists a corresponding single 3D point $X_{i,j}$, where $X \in \mathbb{R}^{W \times H \times 3}$ is a *pointmap*. Given camera intrinsics K and a depthmap D , we can compute $X_{i,j} = K^{-1}[iD_{i,j}, jD_{i,j}, D_{i,j}]$ in the camera coordinate system. In the following, we denote as $X^{n,m}$ the pointmap of image I^n expressed in the coordinate system of camera I^m . To swap the coordinate system from camera I^k to camera I^m , we have $X^{n,k} = P_{m,k}X^{n,m}$ where $P_{m,k} = P_kP_m^{-1}$.

3.1. Overall architecture

We follow the recently proposed DUST3R framework [75]. DUST3R is a breakthrough foundation model based on vision transformers (ViT) that can regress two 3D pointmaps $X^{1,1}, X^{2,1}$ given solely two unposed and uncalibrated input images. Nevertheless, DUST3R’s inability to leverage potentially available information about cameras or depth limits its practical use. To remedy these drawbacks, we enhance the original DUST3R network $\mathcal{F}$ in two ways.

First, we introduce specific modules to incorporate any subset of extra information such as camera intrinsics, camera poses and depthmaps seamlessly into DUST3R. Second, we predict an additional pointmap $X^{2,2}$, which represents the pointmap of image I^2 in its own coordinate system. Predicting three pointmaps offers further capabilities, namely the possibility of recovering all information about *both* cameras in a single forward pass, as explained in Sec. 3.3. We first describe the overall architecture, see Fig. 2, and then detail the dedicated modules to inject auxiliary information.

Encoder. We use a shared ViT encoder to encode both images independently. In addition to I , the encoder can receive auxiliary information about intrinsics K and depth D for each image. For the two input images I^1, I^2 and their respective auxiliary information $\Omega_1 \in \sigma(\{K_1, D_1\}), \Omega_2 \in \sigma(\{K_2, D_2\})$, where σ denotes the set of all subsets, the encoder processes the information in a Siamese manner:

$$F^1 = \text{Encoder}(I^1, \Omega_1), \quad F^2 = \text{Encoder}(I^2, \Omega_2). \quad (1)$$

Decoder. Likewise, the network has two decoders, each with its corresponding head, one predicting $X^{1,1}$ and the other one estimating $X^{2,1}$ and $X^{2,2}$. Both decoders communicate via cross-attention between their own tokens and the outputs of the previous block of the other decoder. Each decoder may receive the relative pose P_{12} as additional input or not. Assume we provide the auxiliary information $\Omega_D \in \sigma(\{P_{12}\})$ at the i -th block of both decoders:

$$\begin{aligned} G_i^1 &= \text{DecoderBlock}_i^1(G_{i-1}^1, G_{i-1}^2, \Omega_D), \\ G_i^2 &= \text{DecoderBlock}_i^2(G_{i-1}^2, G_{i-1}^1, \Omega_D), \end{aligned} \quad (2)$$

After B decoder blocks in each branch, the head regresses

several pointmaps and their associated confidence map:

$$\begin{aligned} X^{1,1}, C^{1,1} &= \text{Head}^1(G_B^1), \\ X^{2,1}, X^{2,2}, C^{2,1}, C^{2,2} &= \text{Head}^2(G_B^2). \end{aligned} \quad (3)$$

Unlike DUST3R which uses a DPT head [50], here we employ a linear head for both Head^1 and Head^2 as it is more efficient and performs similarly.

3D Regression Loss. Following DUST3R, we supervise the model to minimize the distance between ground-truth and predicted pointmaps in a scale-invariant manner, allowing the model to train on multiple datasets with various scales. The regression loss between predicted and ground-truth pointmaps (resp. $X^{n,m}$ and $\hat{X}^{n,m}$) at pixel (i, j) is defined as $\mathcal{L}_{i,j}^{\text{regr}}(n, m) = \|X_{i,j}^{n,m}/z_m - \hat{X}_{i,j}^{n,m}/\hat{z}_m\|$, where $z_m, \hat{z}_m$ serve as scale normalizer. That is, z_m is the average norm of all valid 3D points expressed in coordinate system of image I^m , i.e. $z_1 = \text{norm}(X^{1,1} \cup X^{2,1})$, $z_2 = \text{norm}(X^{2,2})$ and likewise for $\hat{z}_1, \hat{z}_2$, with $\text{norm}(X) = \text{mean}(\{\|X_{i,j}\| \mid i, j \in \mathcal{D}_X\})$ and $\mathcal{D}_X$ the set of valid pixels.

Confidence-aware loss. The model jointly learns to predict a confidence level $C_{i,j}^{n,m}$ of each pixel (i, j) . The confidence-aware regression loss for a given pointmap $X^{n,m}$ can be expressed as the 3D regression loss $\mathcal{L}^{\text{regr}}$ weighted by the confidence map:

$$\mathcal{L}^{\text{conf}}(n, m) = \sum_{i,j \in \mathcal{D}} C_{i,j}^{n,m} \mathcal{L}_{i,j}^{\text{regr}}(n, m) - \alpha \log C_{i,j}^{n,m}, \quad (4)$$

with $\alpha = 0.2$. This loss penalizes the model less when the prediction is not accurate on harder areas, encouraging the model to extrapolate. The final loss is expressed as

$$\mathcal{L} = \mathcal{L}^{\text{conf}}(1, 1) + \mathcal{L}^{\text{conf}}(2, 1) + \beta \mathcal{L}^{\text{conf}}(2, 2), \quad (5)$$

where β is a hyper-parameter typically set to $\beta = 1$.

3.2. Adding Versatile Conditioning

The knowledge of auxiliary information can significantly enhance 3D predictions at test time. Our model leverages up to five different modalities, which include two intrinsics K_1, K_2 , two depthmaps D^1, D^2 for each image and the relative pose P_{12} . To condition the output on it, we first embed the auxiliary information using dedicated MLPs and then inject these embeddings at different points in the pipeline.

We discuss different options in this regard. In the first strategy, denoted as ‘embed’, we simply add the auxiliary embeddings to the token embeddings before the first transformer block, as would be done for standard positional embeddings [16]. In the second strategy, denoted as ‘inject- n ’, we instead insert dedicated MLPs for each modality in a subset of n transformer blocks, as shown in Fig. 3. Our experiments show that ‘inject-1’ performs slightly better than ‘embed’ and similarly with ‘inject- n ’, where $n > 1$, see

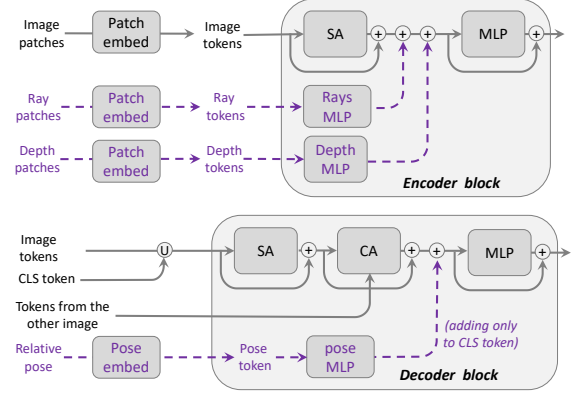


Figure 3. **Architecture for injecting auxiliary modalities.** *Top:* injection of optional intrinsics and depth into the encoder. Intrinsics are encoded into ray patches, sparse depth is patchified. Each of these modalities goes into a block-specific MLP and are token-wise added in the middle of the transformer block. *Bottom:* injection of optional relative pose into the decoder. The relative pose is fed to a first embedding layer followed by an MLP. This token is added to the CLS token of the decoder after the self-attention (SA) and cross-attention (CA), but before the MLP. Experiments show that injection in the first block only suffices.

Sec. 4.5. We now describe how to compute the initial embeddings for each specific modality.

Intrinsics. Following [89], we generate camera rays from the intrinsic matrix $K \in \mathbb{R}^{3 \times 3}$, thereby establishing a direct correspondence between RGB pixels and rays. The ray at pixel location (i, j) is computed as $K^{-1}[i, j, 1]$ and encodes the viewing direction of that pixel w.r.t. the current camera pose. This allows us to potentially process non-centered crops and hence to perform inference in higher image resolutions, see Sec. 3.3. Similarly to RGB inputs, we patchify and embed dense rays and provide them to the encoder.

Depthmaps / Point Clouds. Given a depthmap D and its sparsity mask M , we first normalize $D' = D/\text{norm}(D)$ to handle any depth ranges at train and test time. As for RGB and rays, we patchify the stacked maps $[D', M] \in \mathbb{R}^{W \times H \times 2}$ and compute patch embeddings, which are then fed to the encoder. By jointly patchifying the depth and its valid mask we can handle any level of sparsity.

Camera Pose. Given the relative pose $P_{12} = [R_{12}|t_{12}]$, we normalize the translation scale as $t'_{12} = t_{12}/\|t_{12}\|$ since our output is unscaled. Unlike depthmaps or camera intrinsics, the camera pose cannot be expressed as a dense pixel map. Rather, camera pose affects the whole pixels between two images, so the embedding is instead added to the global CLS token of both decoders.

3.3. Downstream Tasks

Depthmaps. In the pointmap representation, the z -axis of $X^{1,1}, X^{2,2}$ directly corresponds to the depthmaps of the first and second image, respectively. Note that DUST3R

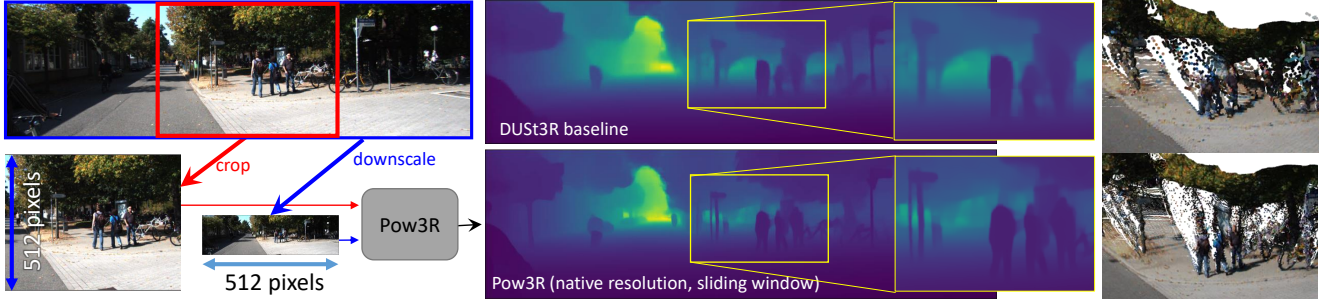


Figure 4. **Native resolution example.** Methods like DUST3R require centered principal point and would thus downscale the image to its training resolution, leading to blurry prediction (top). With camera intrinsics input, we can process any crop in the image leading to native resolution prediction (bottom row), for example by dividing the image into 4 crops, that are independently process as the first input image, while the second input image of the network is set to the downscaled image.

only outputs $X^{1,1}$, hence it requires two forward passes of the pairs (I^1, I^2) and (I^2, I^1) to extract both depthmaps.

High-resolution processing. A limitation of DUST3R is that it is trained for a specific resolution and does not generalize to larger resolutions. A simple fix would be to include higher resolutions during training, but (i) many datasets do not provide high-resolution ground-truths, and (ii) the training cost could become prohibitive. Pow3R, in contrast, can handle crops natively given camera intrinsics of the crop, as these provide the crop position information (*i.e.* via focal length and principal point). We can thus perform prediction in a sliding window fashion, yielding predictions matching any target resolution by simple stitching. Comparison between standard and higher resolution processing are shown in Fig. 4. Note that prediction for each crop may have a different scale, by design, and cannot be stitched directly. We find that simply computing the median scale factor in overlapping areas is sufficient, and we blend the overlapping regions based on confidence without further post-processing.

Focal Estimation. Similar to DUST3R, we can recover focals for both input images from pointmap $X^{1,1}$ and $X^{2,2}$ with the robust Weiszfeld fast iterative solver [47]. Similar to depthmap prediction, DUST3R needs to infer (I_2, I_1) in order to compute the focal of the second image, while Pow3R predicts them in a single pass.

Relative Pose Estimation. Pow3R predicts the relative pose directly by Procrustes alignment [8, 37] to get the scaled relative pose $P^* = [R^* | t^*]$ between $X^{2,2}$ and $X^{2,1}$, as it predicts the pointmaps of the second image in two different camera coordinates.

$$R^*, t^* = \arg \min_{\sigma, R, t} \sum_{i,j} \sqrt{C_{i,j}^{2,2} C_{i,j}^{2,1}} \|\sigma(RX_{i,j}^{2,2} + t) - X_{i,j}^{2,1}\|^2. \quad (6)$$

Procrustes alignment is known to be sensitive to noise and outliers; however, in our case, it performs similar to RANSAC [17] with PnP [22, 29], while being an order of magnitude faster (see Sec. 4.4).

Global alignment. The network $\mathcal{F}$ predicts pointmaps for image pairs. To align all predictions in the same world coor-

dinate system, we resort to the global alignment algorithm from DUST3R [75] that minimizes a global energy function to find per-camera intrinsics, depthmaps and poses that are consistent with all the pairwise predictions. Results of the optimization are global scene point-clouds, which can for instance serve for multi-view stereo estimation.

3.4. Training Procedure

During training, we feed the network with annotated image associated with random subsets of auxiliary information, the goal being for it to learn to handle any situation at test time. For each pair, we first chose a random number m of modalities with uniform probability, and then randomly select the m modalities likewise. Depthmaps are randomly sparsified. When giving intrinsics, we also perform aggressive non-centered cropping with 50% probability, so that the network learns to perform high-resolution inference.

Training data and recipe. We follow the training recipe from DUST3R [75] in terms of training data and protocol. Namely, we use the provided pairs for 8 datasets: Habitat [58], MegaDepth [32], ARKitScenes [6], Static Scenes 3D [41], BlendedMVS [82], ScanNet++ [84], Co3D-v2 [51] and Waymo [67], resulting in a total of 8.5M image pairs. These datasets include indoor and outdoor scenes, as well as real and synthetic ones. To train the model, we follow [75] and first train the network with a resolution of 224px for 3 days on 8 A100 GPUs, and then finetune it in 512px with variable aspect-ratio for 2 more days.

4. Experimental Evaluation

We first study the impact of auxiliary information guidance (Sec. 4.1). We then experiment across many 3D vision tasks (Secs. 4.2 to 4.4) without retraining, *i.e.* mostly zero-shot settings. We finally present some ablations in Sec. 4.5.

DUST3R baseline. We point out that our overall training data, protocols and architecture are exactly the same as DUST3R, except for the additional head that predict $X^{2,2}$ (+0.1% parameters) and for the modules used for injecting

	aux. modalities					focal	depth	rel. pose	
	K1	K2	D1	D2	RT	acc@1.015	τ @1.03	RRA@2°	RTA@2°
DUST3R	×	×	×	×	×	36.6	77.6	69.2	49.7
Pow3R	×	×	×	×	×	39.4	79.4	71.9	53.8
	✓	×	×	×	×	75.4(+36.0)	80.6 (+1.2)	77.1 (+5.3)	62.8 (+8.9)
	×	✓	×	×	×	67.3(+27.9)	80.3 (+0.9)	74.5 (+2.6)	59.1 (+5.2)
	✓	✓	×	×	×	98.0(+58.6)	81.4 (+2.0)	80.3 (+8.5)	74.2(+20.3)
	×	×	✓	×	×	48.9 (+9.5)	89.1 (+9.7)	82.5(+10.6)	64.9(+11.0)
	×	×	×	✓	×	49.5(+10.1)	91.0(+11.7)	83.4(+11.5)	64.8(+10.9)
	×	×	×	×	✓	58.2(+18.9)	95.4(+16.0)	89.6(+17.7)	77.6(+23.8)
	×	×	×	×	✓	48.6 (+9.2)	81.6 (+2.2)	92.3(+20.5)	77.0(+23.1)
	✓	✓	✓	✓	×	99.2(+59.8)	95.4(+16.0)	95.7(+23.9)	94.3(+40.5)
	✓	✓	×	×	✓	98.1(+58.7)	82.9 (+3.6)	95.1(+23.2)	87.3(+33.5)
	×	×	✓	✓	✓	68.6(+29.2)	95.4(+16.0)	98.1(+26.2)	91.3(+37.5)
	✓	✓	✓	✓	✓	99.3(+59.9)	95.4(+16.0)	99.0(+27.2)	98.1(+44.3)

Table 1. **Impact of guiding at test time** for models trained at 224×224 resolution. We report performances on Habitat for DUST3R, which cannot handle auxiliary modalities, and our single model with different sets of modalities; we show in green the absolute improvement w.r.t. the results without auxiliary modality.

the auxiliary information (+4% parameters for the inject-1 variant that we use unless stated otherwise).

Metrics. When evaluating focal lengths, we report the accuracy for a tight relative error threshold of 1.5%. For evaluating depthmap, we report the Absolute Relative Error (rel) and the Inlier Ratio (τ) at a threshold of 3%. When evaluating poses, we opt for scale-invariant metrics and measure the relative rotation and translation accuracies (RRA and RTA resp.) of the relative pose at task-specific thresholds, and the mean Average Accuracy (mAA), defined as the area under the curve accuracy of the angular differences at min(RRA, RTA).

4.1. Guiding the output prediction

Impact of guidance. We first study the impact of providing different subsets of auxiliary information in terms of downstream 3D vision performance. Namely, we report accuracies in Tab. 1 for focal, depth, and rotation prediction on the Habitat validation dataset. In the supplementary material, we also provide results on the Infinigen dataset [49], which is not included in our training set, and observe similar outcomes. We first note that in the absence of modalities, Pow3R performs slightly better than a DUST3R baseline, even though Pow3R is trained with just RGB images during a fraction of its training. This shows that Pow3R does not lose its native strength for unconstrained regression despite having gained more capabilities.

Interestingly, we observe in Tab. 1 the impact of each modality for each prediction task, and the synergetic effects of interacting modalities. For instance, the model performs better for focal length estimation with depthmaps and/or relative pose, and Pow3R’s depth prediction improves with intrinsics and/or relative pose. Moreover, relative pose prediction becomes more accurate with intrinsics and/or depthmap. Overall, the model’s performance consistently improves when provided with additional camera or scene priors, demonstrating the effectiveness of Pow3Rin

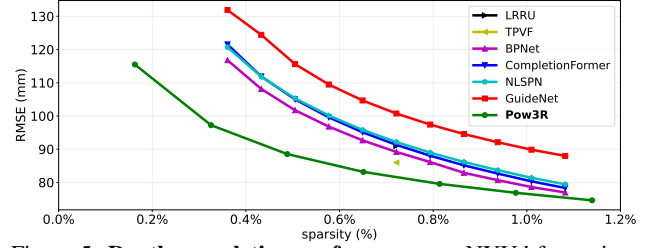


Figure 5. **Depth completion performance** on NYUd for various input depth sparsity.

	aux. mod.		High-Res.	KITTI		T&T		Time (sec)↓
	Ks	RT		rel↓	τ ↑	rel↓	τ ↑	
DUST3R512	×	×	×	5.4	49.5	3.3	75.1	0.13
Pow3R512	✓	✓	×	5.3	48.7	3.2	78.2	0.13
Pow3R512	✓	✓	(n)	7.5	34.4	3.9	68.0	0.48
Pow3R512	✓	✓	✓	4.6	53.5	2.5	82.3	0.40

Table 2. **Multi-view depth estimation results in high-resolution** using a sliding-window scheme and providing the crop intrinsics to Pow3R. (n): naively inputting high-resolution images without downsampling to Pow3R.

handling any combination of priors in a unified fashion. We are not aware of any other approach that can seamlessly integrate all these priors in a learning-based method.

Depth completion. We also evaluate Pow3R on the depth completion task given RGB images with sparse depthmaps. We vary the sparsity (the ratio of pixels with known depth) and demonstrate its impact on the NYUv2 dataset [63] in Figure 5. Although Pow3R is not trained on the NYUv2 dataset, it still achieves state-of-the-art results for sparse depth completion. Our analysis indicates that the true performance of Pow3R on NYUv2 is likely underestimated due to numerous errors in the ground-truth depth annotations; further details are available in the supplementary material. On the KITTI [20] dataset, we obtain a mean absolute error below 30cm, which is competitive with the state of the art, considering that KITTI is not included in the training set.

High-Resolution processing. With the possibility of specifying camera intrinsics, new capabilities arise. Namely, Pow3R can perform high-resolution pointmap regression with a simple sliding-window approach. We process fixed-resolution crops of the original images under the guidance of both (i) the crop information, provided in term of camera intrinsics, and (ii) a second image, itself possibly down-scaled or cropped as well. More details are provided in Sec. 3.3 and Fig. 4. When applied, results significantly improve compared to processing the images at 512-pixel resolution, as shown in Tab. 2. We also compare with a naive baseline where we simply feed high-resolution images directly to Pow3R, but it dramatically fails.

Camera controllability. We study how *controllable* are the regression outputs on 10K Habitat test pairs given arbitrary camera parameters. For each test pair with ground-truth fo-

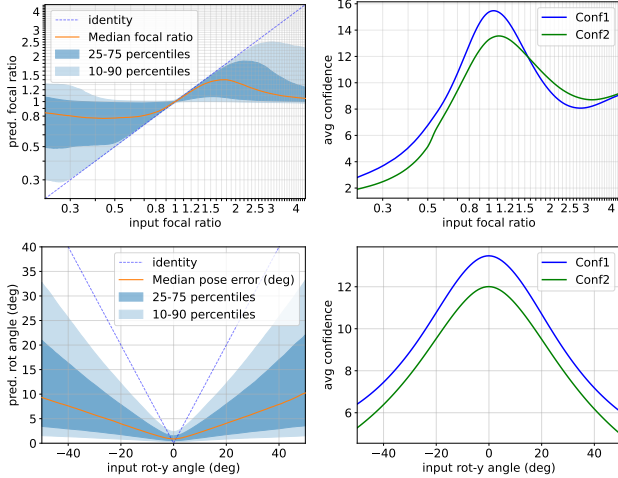


Figure 6. **Controllability study.** We study how the model reacts when presented with false focal (top) and false relative pose (bottom). We plot the input vs. predicted camera parameters in the left column and the average confidence in the right column. Interestingly, the model seems to not take the input information at face value and instead makes an informed decision whether to use it or not, in which latter case the confidence becomes low.

cal f and relative pose P_{12} , we purposefully input a *wrong* focal (resp. relative pose) and measure the output focal $\hat{f}$ (resp. pose $\hat{P}_{12}$) as predicted from the raw pointmaps. Results are presented in Fig. 6 in terms of focal ratio $\hat{f}/f$ and angle $d_{\text{geodesic}}(P_{12}, \hat{P}_{12})$. Interestingly, the model accepts to follow the guidance whenever it is close to the ground truth value (*i.e.* staying close to the identity transformation) but then refuses to follow it when it gets too different, showing that the model is able to somehow ‘judge’ of the guidance’s worth. In all cases, the output confidence clearly correlates with the agreement between the guidance and the ground truth. Overall, the model is thus not taking the auxiliary information at face value, and instead combines it with the other modalities (*e.g.* RGB) in a more subtle and informed way than what could be achieved with handcrafted methods. This suggests that Pow3R would be robust to noise in the auxiliary information.

4.2. Multi-View Depth Estimation

Following RobustMVD [62], performances are measured on the KITTI [20], ScanNet [14], ETH3D [61], DTU [1], and Tanks and Temples [28] datasets. We report in Tab. 3 the Absolute Relative Error (rel) and Inlier Ratio (τ) for each dataset when using (1) images alone, (2) images with relative poses, (3) images with camera intrinsics, and (4) images with both modalities, and provide results for more state-of-the-art methods in the supplementary material.

Results. Pow3R again performs roughly on par with DUST3R in the absence of auxiliary information (*e.g.* 3.73 average error across all datasets for DUST3R-512 vs. 3.64

Method	GT intrinsic	Co3Dv2			RealEstate10K	Speed (fps)
		RRA@15	RTA@15	mAA(30)	mAA(30)	
Colmap+SG [56]	✓	36.1	27.3	25.3	45.2	-
PixSfM [34]	✓	33.7	32.9	30.1	49.4	-
RelPose [88]	×	57.1	-	-	-	-
(a) PoseDiff [73]	×	80.5	79.8	66.5	48.0	-
RelPose++ [33]	×	85.5	-	-	-	-
RayDiff [89]	×	93.3	-	-	-	-
DUST3R (PnP) [75]	×	94.3	88.4	77.2	61.7	3.2
(b) Pow3R (PnP)	×	94.8	89.9	78.5	62.5	3.2
Pow3R (Pro)	×	94.6	90.3	78.1	66.3	30.9
Pow3R w/ K (Pro)	✓	95.0	92.1	82.2	72.5	30.1

Table 5. **Multiview pose estimation on the CO3Dv2 and RealEstate10K.** (a) denotes multiview methods, while (b) denotes pairwise methods. Pow3R with Procrustes alignment (Pro) performs better than DUST3R (PnP) and other multi-view based pose estimation approaches, and even more so with known intrinsics, while being much faster. RealEstate10K is not in the training set.

for Pow3R-512). However, it is clear that providing auxiliary information consistently improves the depth estimates. In fact, Pow3R-512 with both pose and intrinsics achieves state-of-the-art performance on the ETH3D, DTU, and Tanks and Temples datasets, as well as on average across datasets. Interestingly, intrinsics tend to improve performance more than the GT pose, which alone seems to have a negligible impact. This might be due to the fact that camera pose primarily helps the prediction of $X^{2,1}$, while the depthmap of this task is obtained from the z axis of $X^{1,1}$ alone. Still, we observe a clear synergy in providing jointly intrinsics *and* relative pose.

4.3. Multi-View Stereo

We evaluate the quality of global 3D reconstructions on the DTU [1] benchmark in the presence and absence of auxiliary information. For each subset of available information from $\{(K_1, K_2), P_{12}\}$, we extract pairwise predictions from image pairs and align them in a global coordinate system, as described in Sec 3.3. Performance is evaluated in terms of accuracy, which is the smallest Euclidean distance to the ground-truth, and completeness as the smallest distance to the reconstructed shape, with the overall average.

Results are presented in Tab. 4. We first attempt at reproducing the original DUST3R results using the official codebase and obtain slightly worse results than reported in [75]. Using the same codebase, Pow3R with RGB only is able to improve over these and match the original results. Interestingly, adding ground-truth intrinsics results in a significant boost (19% rel. improvement), but adding the camera poses instead slightly *degrades* the results (-4%). However, using both intrinsics and pose results in a major improvement (36% rel. improvement), demonstrating a synergetic effect between both modalities. We hypothesize that the pose is of little value for easy cases like DTU, and that it does not help the model to properly resolve the focals.

Method	GT range	KITTI		ScanNet		ETH3D		DTU		T&T		Average	
		rel↓	τ ↑	rel↓	τ ↑	rel↓	τ ↑	rel↓	τ ↑	rel↓	τ ↑	rel↓	τ ↑
COLMAP [59, 60] (K+RT)	×	12.0	58.2	14.6	34.2	16.4	55.1	0.7	96.5	2.7	95.0	9.3	67.8
COLMAP Dense [59, 60] (K+RT)	×	26.9	52.7	38.0	22.5	89.8	23.2	20.8	69.3	25.7	76.4	40.2	48.8
MVSNet [81] (K+RT)	✓	18.6	30.7	22.7	20.9	21.6	35.6	(1.8)	(86.7)	6.5	74.6	14.2	49.7
Vis-MVSNet [86] (K+RT)	✓	9.5	<u>55.4</u>	8.9	33.5	10.8	43.3	(1.8)	(87.4)	4.1	<u>87.2</u>	7.0	61.4
MVS-Former++ [9] (K+RT)	✓	29.2	15.2	15.2	21.9	21.4	32.5	(1.2)	(91.9)	7.6	71.5	14.9	46.6
CER-MVS [40] (K+RT)	×	14.3	32.2	21.1	24.3	11.7	47.5	4.1	71.3	6.4	82.1	11.5	51.5
DUST3R [75]	×	5.4	49.5	(3.1)	(71.8)	3.0	<u>76.0</u>	3.9	68.6	3.3	75.1	3.7	68.2
Pow3R	×	5.7	45.7	(3.2)	(68.8)	3.0	74.7	3.0	74.3	3.3	76.6	3.6	68.0
Pow3R w/ RT	×	5.7	45.8	(3.2)	(69.7)	<u>2.9</u>	75.6	3.3	71.6	<u>3.2</u>	77.9	3.7	68.1
Pow3R w/ K	×	5.3	48.3	(3.1)	(70.8)	<u>2.9</u>	<u>76.0</u>	1.6	89.9	<u>3.2</u>	77.3	3.2	<u>72.5</u>
Pow3R w/ K+RT	×	5.3	48.7	(3.1)	<u>(71.4)</u>	2.8	77.1	<u>1.5</u>	<u>91.1</u>	<u>3.2</u>	<u>78.2</u>	3.2	73.3

Table 3. **Multi-view depth evaluation:** Pow3R with both pose and intrinsics performs better than DUST3R. More comparisons with the state of the art can be found in the supplementary material. (Parentheses) denote training on data from the same domain.

4.4. Multi-View Pose estimation

We then evaluate Pow3R for the task of multi-view pose prediction on the Co3Dv2 [51] and RealEstate10K [91] datasets. Co3Dv2 comprises over 37,000 videos across MS-COCO 51 categories with ground-truth camera derived from running COLMAP on 200 frames per video. RealEstate10k contains 80,000 YouTube video clips, containing indoor/outdoor scenes, with camera poses obtained via SLAM followed by Bundle Adjustment. For Co3Dv2, we follow the protocol of [73] and for RealEstate10K, we use its test set of 1,800 videos. For each sequence, we randomly select 10 frames and feed 45 pairs to Pow3R. We report the results in terms of RRA, RTA and mAA in Tab. 5.

Results. Pow3R predicts $X^{2,2}$, allowing to estimate the relative pose between 2 images in a single forward using Procrustes Alignment (see Sec. 3.3). In contrast, DUST3R does not output $X^{2,2}$ and relies on PnP to obtain camera poses. Pow3R with PnP achieves slightly better results than DUST3R with PnP. Using Procrustes alignment yields similar results, while being orders of magnitude faster than RANSAC, with a roughly $10\times$ speed-up overall when combining inference and pose estimation times. Finally, performance improves even further with known intrinsics, significantly outperforming all the state of the art.

4.5. Architecture ablation

We evaluate the impact of adding the $X^{2,2}$ prediction in Tab. 6. For these ablations, models are trained on a subset of Habitat and tested on focal, depth, and relative pose prediction, with results averaged over all combinations of auxiliary modalities. DUST3R serves as a baseline, and DUST3R-ft is finetuned with an additional $X^{2,2}$ head. This output can be used at test time to directly predict focals or relative poses, within a single forward pass, instead of having to forward both the original pair and its swapped version. We also report the performance, with and without predicting $X^{2,2}$ and/or using it. In all cases, predicting $X^{2,2}$ leads to slightly better performance while evaluations can be performed in a single forward pass.

We also explore different ways of conditioning auxil-

Method	Acc.↓	Comp.↓	Overall↓
DUST3R [†] [75]	2.677	0.805	1.741
DUST3R (repr.)	2.191	1.598	1.894
Pow3R	2.116	1.370	1.743
Pow3Rw/ K	1.722	1.119	1.420
Pow3Rw/ Rt	2.205	1.429	1.817
Pow3Rw/ K+RT	1.384	0.846	1.115

Table 4. **MVS results** with the accuracy, completeness and their average (in mm) on the DTU dataset. DUST3R[†] means results from the original paper, and ‘repr.’ means reproduced with the official public codebase and checkpoint.

Method	with $X^{2,2}$	Single Forward	Focal acc@1.015	Depth τ @1.03	RRA@2 (Pro) (PnP)	RTA@2 (Pro) (PnP)	Avg.
DUST3R	×	×	36.8	85.4	69.2 72.4	49.8 57.3	60.5
DUST3R-ft	✓	×	38.9	79.1	71.7 74.4	52.9 59.2	62.7
	✓	✓	38.3	78.5	73.1 74.4	56.3 59.2	63.3
Pow3R	×	×	66.0	85.7	81.9 83.8	68.6 73.3	76.5
	✓	×	67.2	86.1	82.9 84.7	70.0 74.6	77.6
	✓	✓	64.8	85.0	84.2 84.7	73.0 74.6	77.7

Table 6. **Impact of regressing $X^{2,2}$.** Metrics (in %) on Habitat are averaged over all combinations of input modalities. The ‘with $X^{2,2}$ ’ column denotes methods trained to predict it. ‘Single forward’ denotes evaluations of models based on $X^{2,2}$ prediction, *i.e.*, in a single forward pass, instead of forwarding image pairs and their swapped versions. The top part shows the original DUST3R, or finetune (-ft) with $X^{2,2}$ prediction.

Arch.	Extra Params	Focal acc@1.015	Depth τ @1.03	RRA@2 (Pro) (PnP)	RTA@2 (Pro) (PnP)	Avg.
Embed	16.0M	64.7	84.7	84.1 84.5	72.8 73.8	77.4
Inject1	22.6M	64.8	85.0	84.2 84.7	73.0 74.6	77.7
Inject4	42.3M	65.9	85.1	84.8 85.2	73.7 75.4	78.3

Table 7. **Architecture ablation.** Results (in %), on Habitat, are averaged over all combinations of input modalities. We compare the injection architecture for different number of blocks with the ‘embed’ architecture in which the extra signals are added before the transformer blocks. ‘Inject1’ offers a good compromise between performance and number of extra parameters.

iary information in Tab. 7. To incorporate additional priors, Pow3R injects those embeddings into the first block of the encoder or decoder, see Fig. 3. We compare this approach to another design choice, where the model embeds them before going through attention blocks. We have also tried to perform injection in more than one block, without significant improvement while the computational cost increases.

5. Conclusion

We have presented Pow3R, a novel approach to guide large 3D vision models, *e.g.* DUST3R, with camera and scene priors in an implicit manner. Not only does it results in consistent gains whenever auxiliary information is available, but it also unlocks new capabilities such as high-resolution processing and sparse-to-dense depth completion.

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